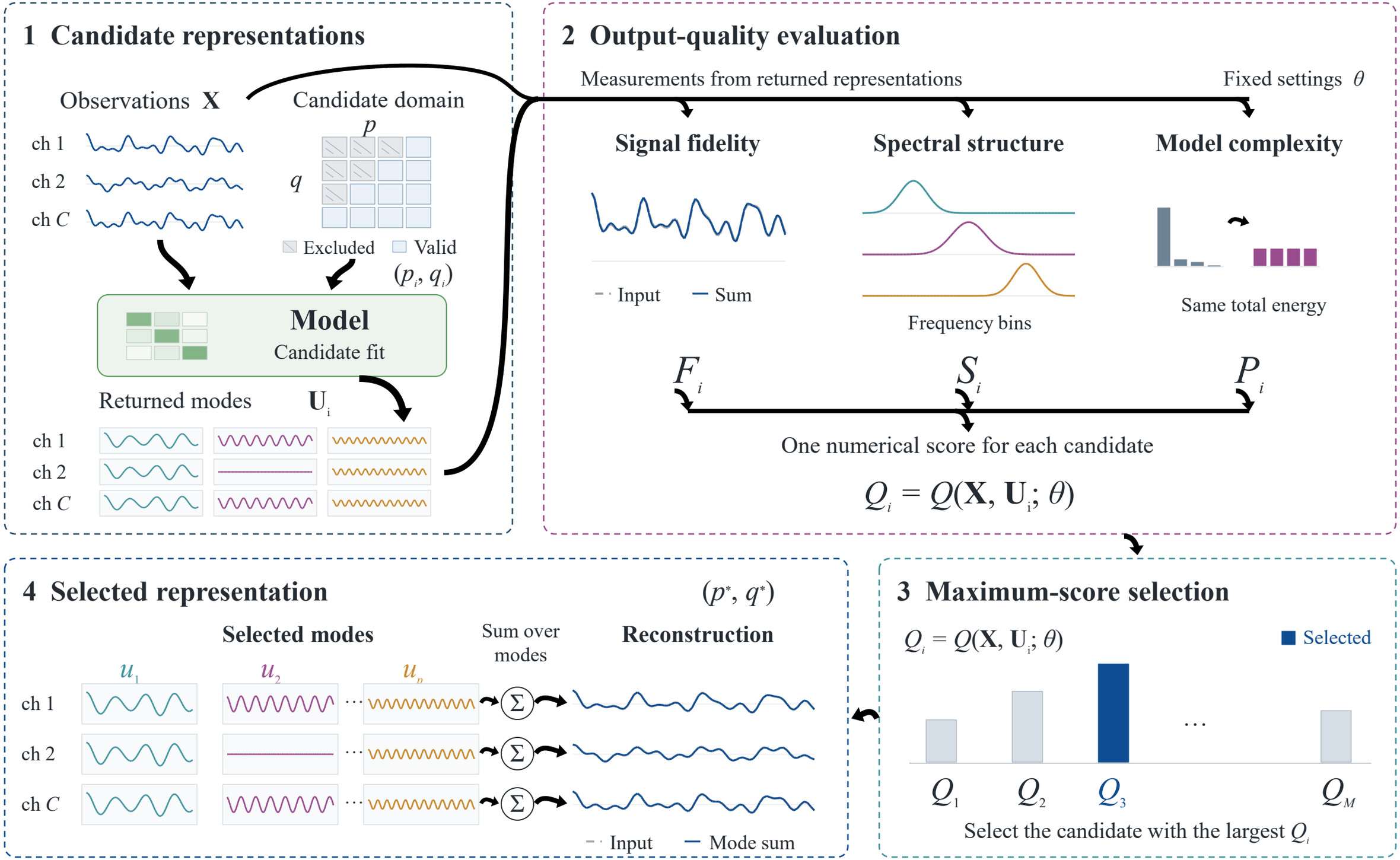

1 Candidate representations
Observations X
ch 1
ch 2
ch C
Candidate domain
p
q
Excluded
Valid
(p_i, q_i)
Model
Candidate fit
Returned modes
U_i
ch 1
ch 2
ch C
2 Output-quality evaluation
Measurements from returned representations
Fixed settings θ
Signal fidelity
Input
Sum
Spectral structure
Frequency bins
Model complexity
Same total energy
F_i
S_i
P_i
One numerical score for each candidate
Q_i = Q(X, U_i; θ)
3 Maximum-score selection
Q_i = Q(X, U_i; θ)
Selected
Q_1
Q_2
Q_3
Q_M
Select the candidate with the largest Q_i
4 Selected representation
(p*, q*)
Selected modes
Sum over modes
Reconstruction
u_1
u_2
u_p
ch 1
ch 2
ch C
Input
Mode sum